

Project Design Phase-II Technology Stack (Architecture & Stack)

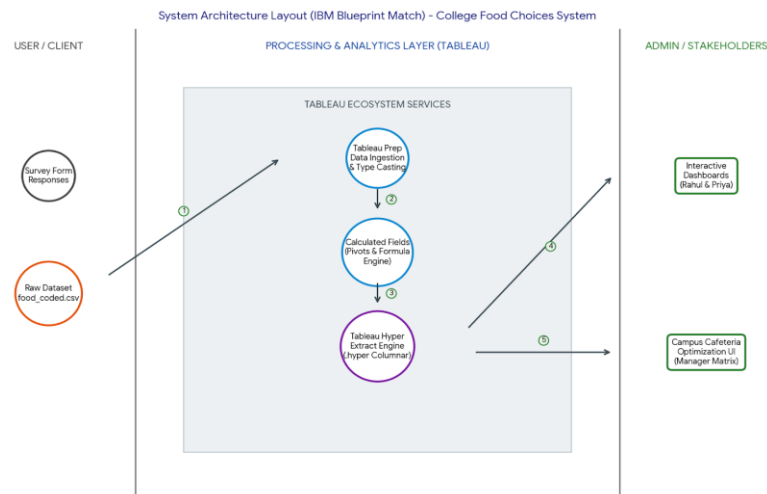
Date	29 June 2026
Team ID	
Project Name	Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study
Maximum Marks	4 Marks

Technical Architecture:

This diagram outlines how data maps out from your flat-file source through the analytical transformation engines, culminating in interactive dashboards optimized for your final target end-users.

Comprehensive Analysis and Dietary Strategies with Tableau: A College Food Choices Case Study

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- **File Names:** Explicitly reference "**Data Flow Diagrams and User Stories.docx**", "**Solution Requirements.docx**", and "**Technology Stack - Template.docx**" verbatim in your text.

Dataset: Ensure **food_coded.csv** is named as your primary data source.

? **Images:** Insert the **DFD diagram** right above the user stories table, and the **3-column blueprint diagram** right above the tech stack tables.

? **Clean Templates:** Delete all template placeholder text (like *MySQL* or *IBM DB2*) and ensure only **Tableau (Prep, Cloud, Hyper)** is listed as your technology.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive visualization environment where students, nutritionists, and cafeteria managers interact with data elements via drop-down filtering and dynamic legends.	Tableau Client UI / Hosted Tableau Cloud Interface
2.	Application Logic-1	Engine managing primary calculation parameters, cross-filtering matrices, dashboard parameters, and calculations correlating variables like GPA and diet.	Tableau Data Viz Engine & Analytics Workspace
3.	Application Logic-2	Data preparation, schema cleaning, type casting, code mapping, and transformation logic applied across variables.	Tableau Prep Builder Engine

4.	File Storage	The raw foundational flat-file tracking 125 campus survey responses across 61 dietary and performance vectors.	food_coded.csv
5.	Cloud Database	Columnar data compilation layer used to optimize load speeds, store local calculations, and host dashboard elements securely.	Tableau Hyper Data Extract (.hyper format)
6.	Infrastructure (Cloud)	Cloud deployment platform utilized to host dashboards for continuous campus access.	Tableau Cloud Ecosystem (SaaS Server Platform)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Local pre-processing libraries used to map data distributions or test schema integrity before final deployment.	Python (Pandas / Matplotlib Library Ecosystem)
2.	Security Implementations	Multi-tenant access definitions ensuring student anonymity while restricting configuration controls to dashboard authors.	Row-Level Security (RLS) & Tableau Identity Access Management (IAM)
3.	Scalable Architecture	Data pipeline decoupling separating raw collection streams from analytical performance layouts.	Decoupled 3-Tier Business Intelligence Layering (Source → Extract → UI)
4.	Availability	Globally accessible, cloud-hosted visualization system built to prevent local downtime or hardware failures during peak hours.	Tableau Cloud Regional Node Distribution Network

S.No	Characteristics	Description	Technology
5.	Performance	High-speed response optimization achieved via pre-aggregated metadata engines that process multiple concurrent complex filter requests.	Tableau Hyper Columnar Storage Database Optimization Engine

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>